



# Visible and Near-Infrared Spectral Effects of Porosity and Albedo on Asteroid and Lunar Analog Samples

Matthew A. Hernandez ([ma040619@ucf.edu](mailto:ma040619@ucf.edu)), A. C. Martin, and K. L. Donaldson Hanna

Department of Physics

University of Central Florida, Orlando, FL

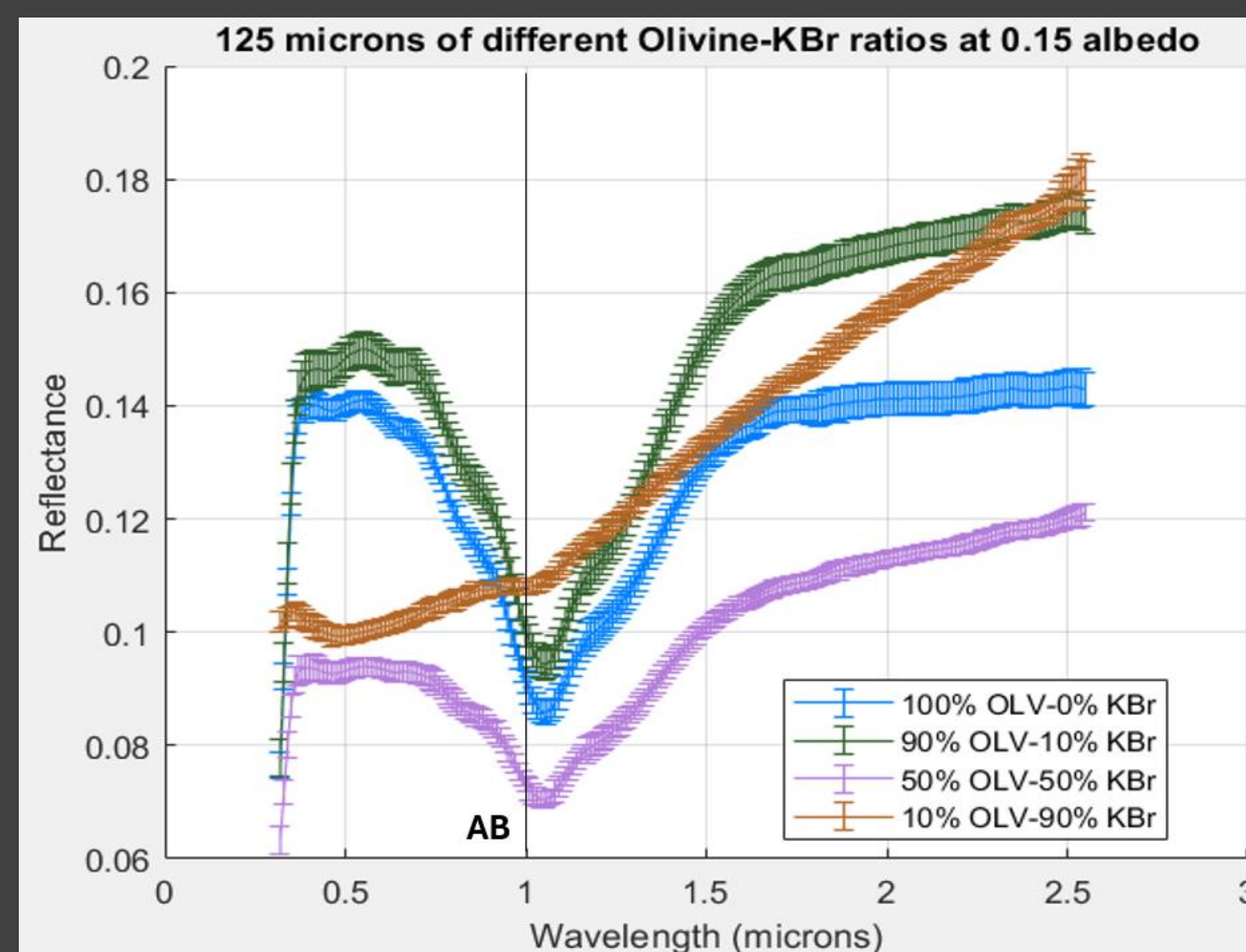
## Introduction

Spectroscopy is a fundamental tool with applications in various scientific fields, particularly in the study of the surfaces of airless bodies like the Moon and asteroids. Lunar regolith simulants are used to simulate known properties of lunar regolith including composition, porosity, albedo, and particle size fraction. **Albedo** is a term astronomers and scientists use to describe the reflectivity of an object in space.

Other uses for the simulants include testing how lunar dust affects various subsystems on lunar landers and spacecrafts. We are currently examining the **Visible and Near-Infrared (VNIR) reflectance** spectra of samples (Figures 2-5), with the intention of analyzing them in the **Mid-Infrared (MIR)** range.

- The VNIR range extends from approximately **0.4  $\mu\text{m}$  to 2.5  $\mu\text{m}$** .
- The MIR range extends from approximately **2.5  $\mu\text{m}$  to 25  $\mu\text{m}$** .

Figure 1: Example Olivine + KBr spectra, all with 0.15 albedos



- Fe-bearing minerals like olivine have diagnostic absorption bands near 1.0  $\mu\text{m}$  indicative of composition and mineral chemistry. Olivine (as seen in **Figure 1**) has 3 overlapping absorption features that create a wide feature centered near 1.0  $\mu\text{m}$ .

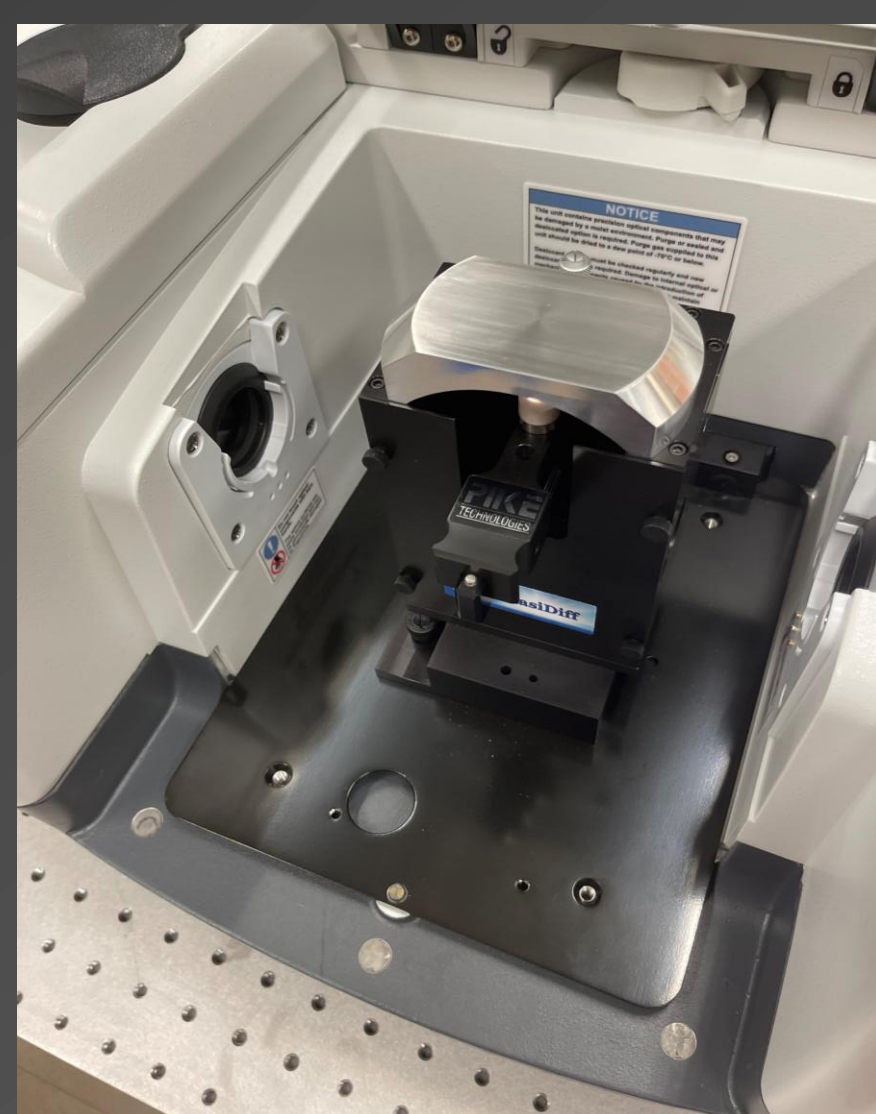


Figure 2: Reflectance accessory inside the FTIR spectrometer

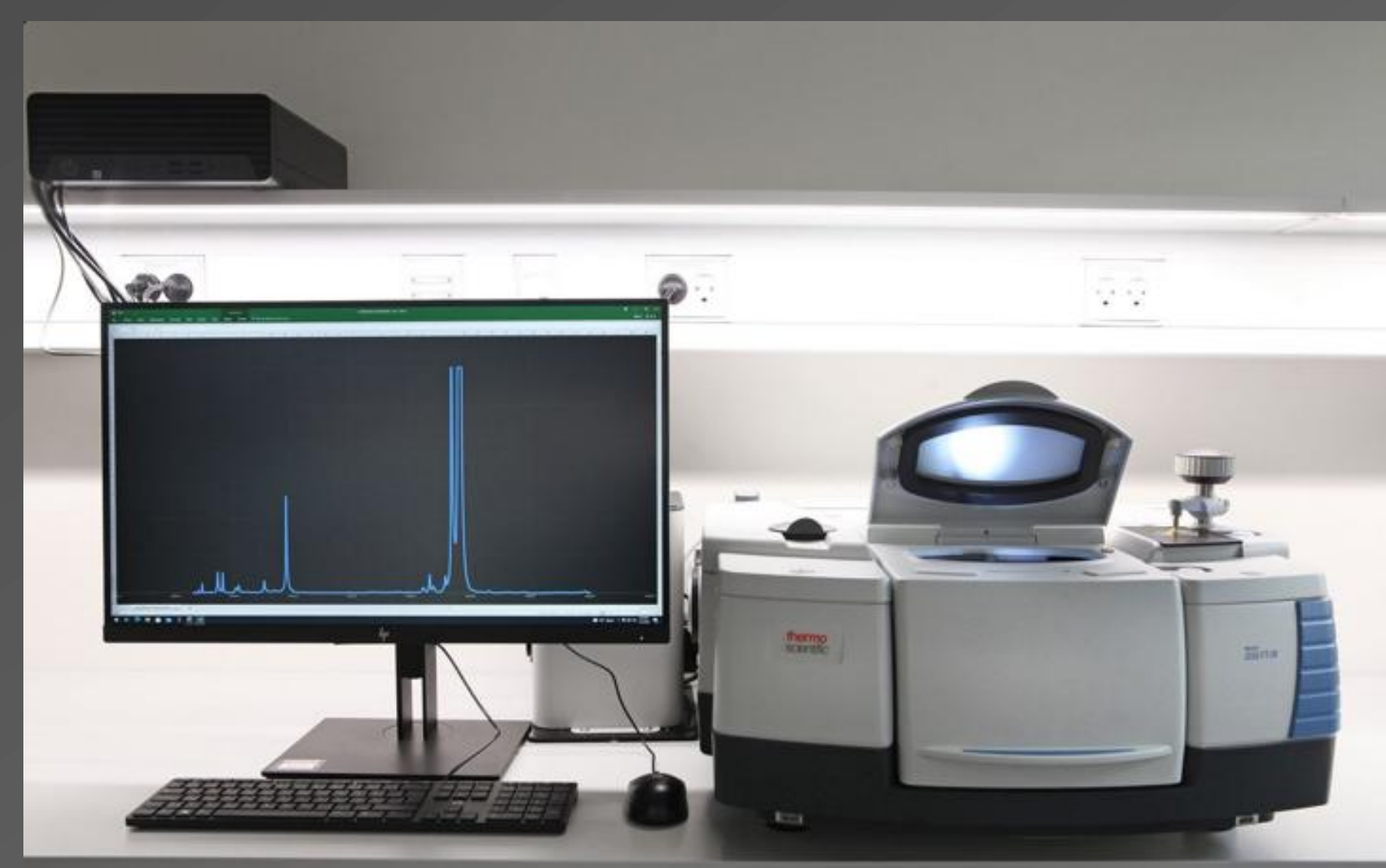


Figure 3: Workstation with computer and FTIR spectrometer

## Sample Measurements & Methods

I produced and examined physical mixtures of San Carlos olivine, potassium bromide (KBr) and carbon black in the particle-size fraction of **45-125 microns ( $\mu\text{m}$ )**. The samples were ground, sieved, and mixed to ratios based on regolith porosity of **0%, 10%, 50%, and 90%**. Carbon black was then added to each sample until a specific albedo was achieved as measured at 0.7 microns.

Figure 4: 100% Olivine + 0% KBr (45-125  $\mu\text{m}$ ).

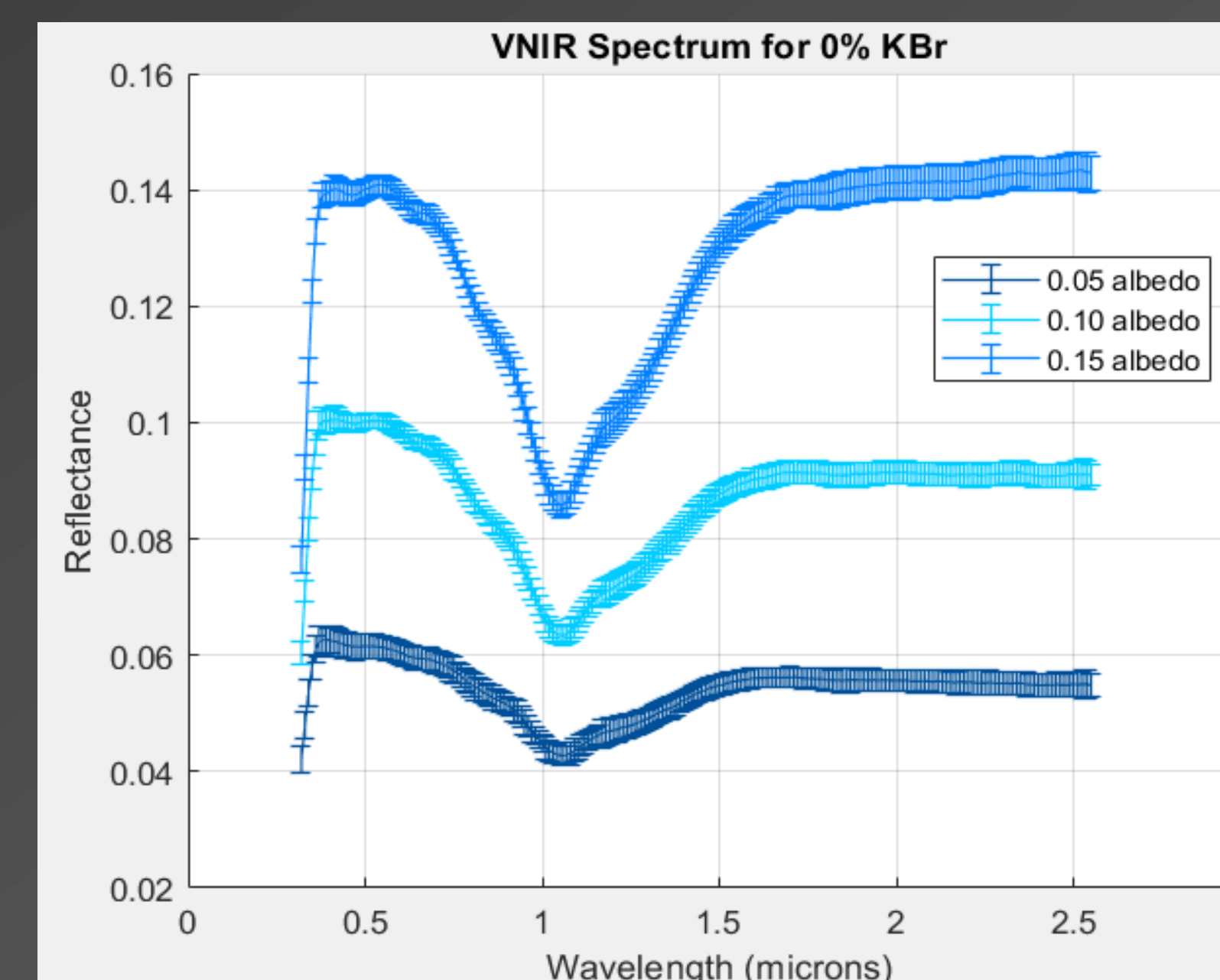


Figure 5: 90% Olivine + 10% KBr (45-125  $\mu\text{m}$ ).

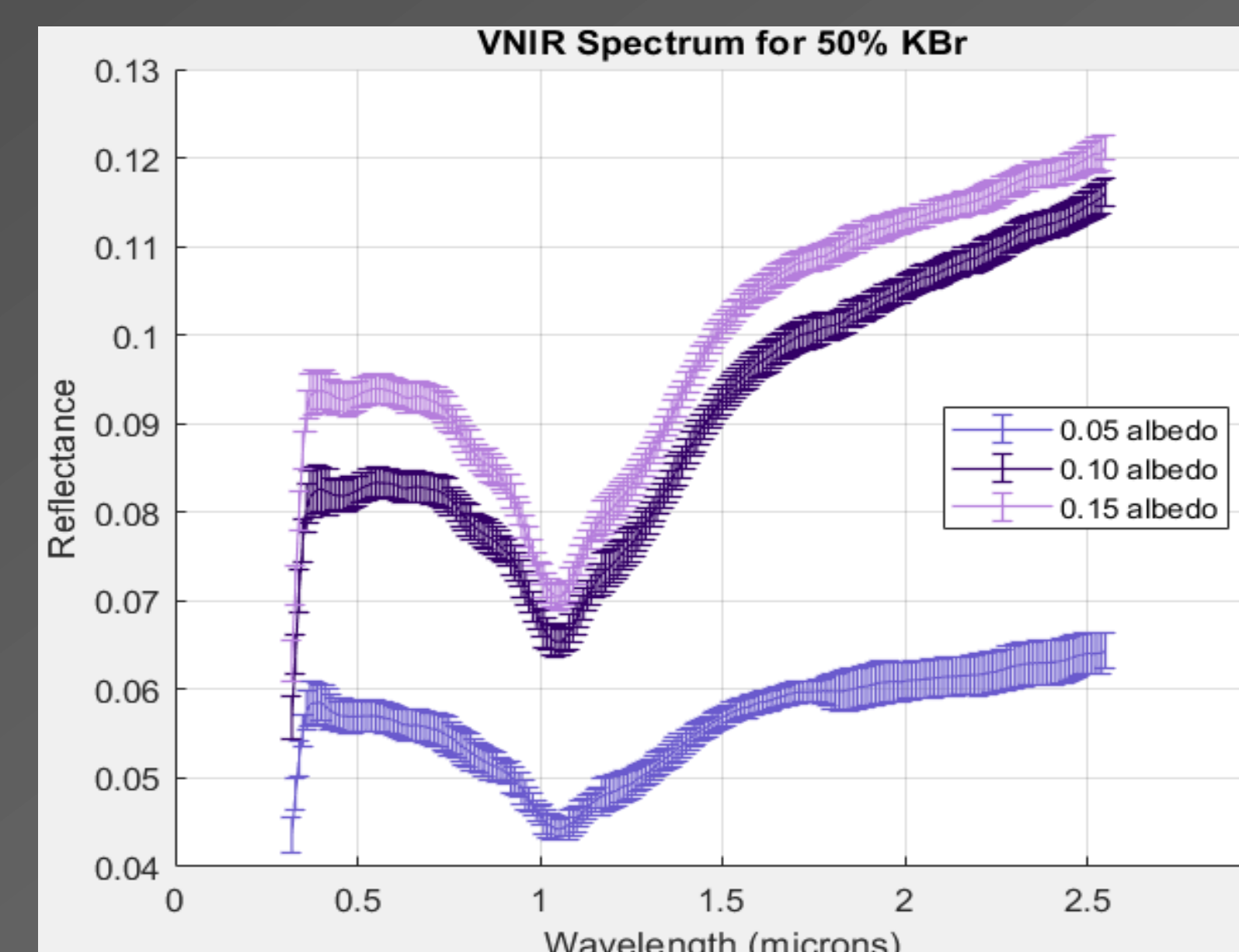
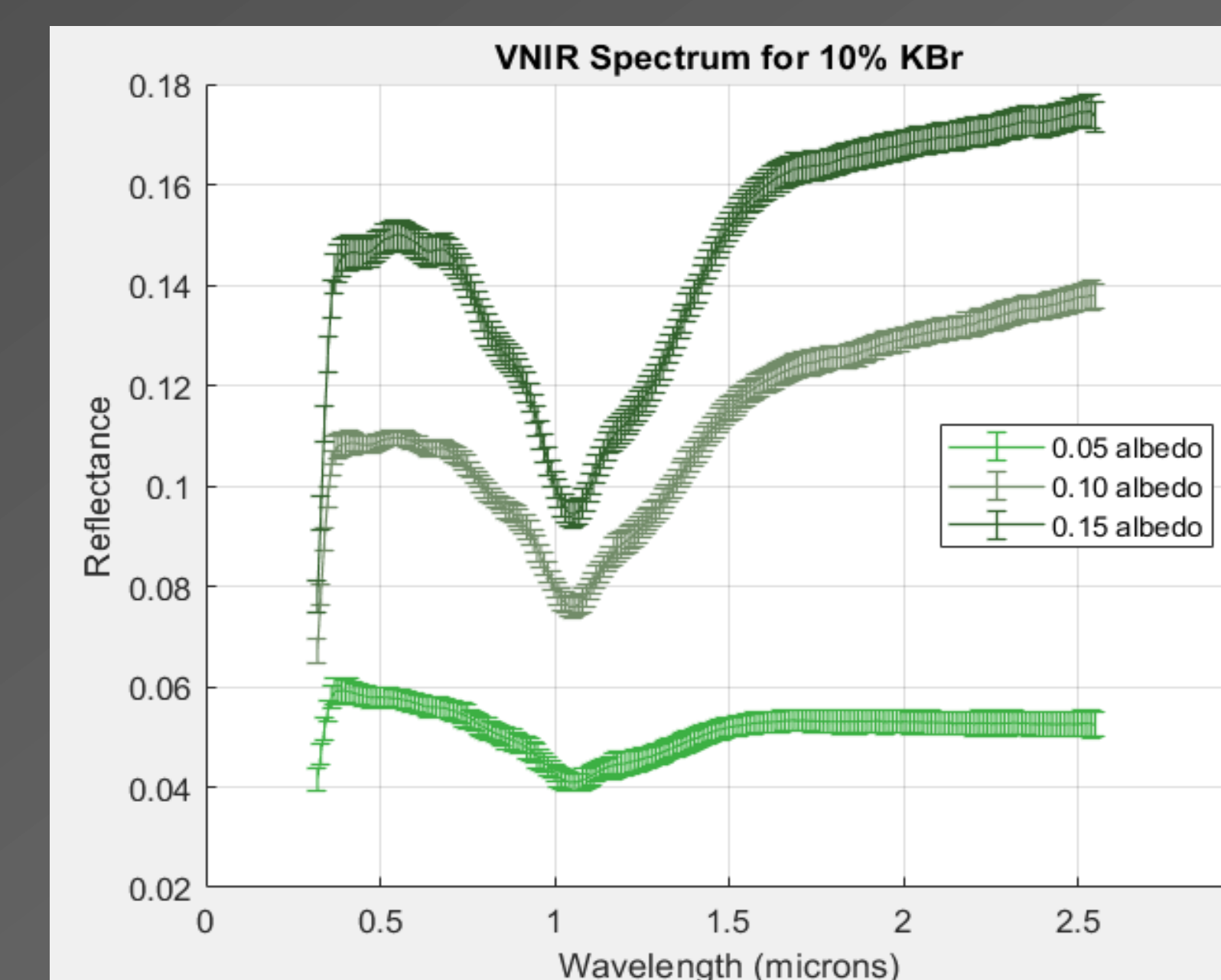


Figure 6: 50% Olivine + 50% KBr (45-125  $\mu\text{m}$ ).

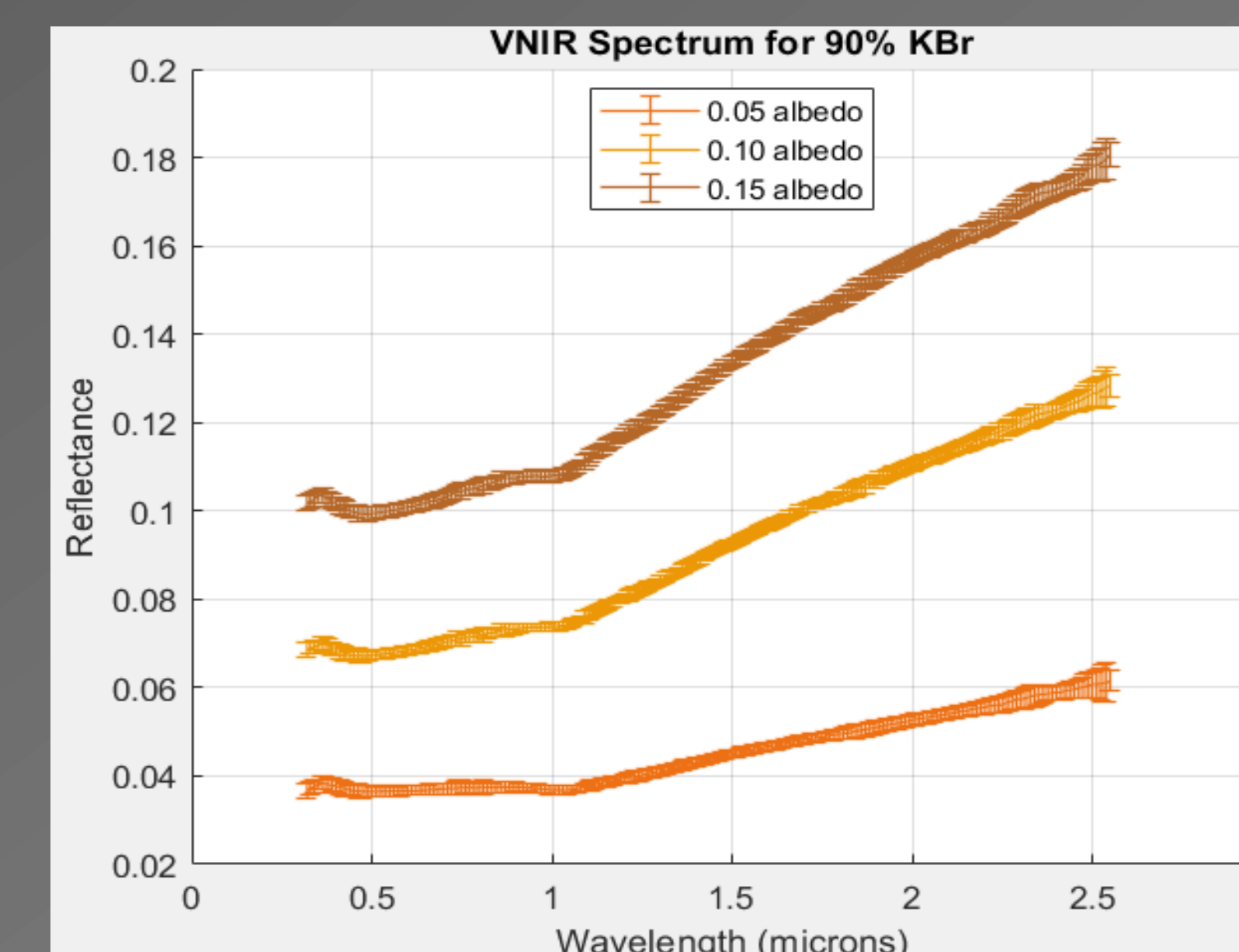


Figure 7: 10% Olivine + 90% KBr (45-125  $\mu\text{m}$ ).

### Methods

I adjusted the albedo of each sample by mixing them with carbon black to achieve albedo values of 0.15, 0.10, and 0.05. Due to the challenges in precisely attaining target albedo values, we allowed for a range of  **$\pm 0.005$** . I measured the albedo of each sample using a Thermo Nicolet iS50 Fourier Transform Infrared (FTIR) spectrometer (**Figure 3**) using a PIKE Diffuse Reflectance Accessory as shown in **Figure 2**.

In Figures 4-7, you'll find spectra plots depicting different ratios of Olivine to KBr. The legend identifies each plot's respective target albedo.

## Results & Future Work

I found that the spectra containing **0%** (**Figure 4**) and **10%** (**Figure 5**) KBr exhibit similar spectral shape, while the spectra for **90%** (**Figure 5**) KBr have much steeper slopes and reduced spectral contrast than other spectra. Another upcoming task involves generating spectral graphs in MIR using collected data, where we will analyze key spectral features within the MIR spectra.

Following this, I will develop MATLAB and Python code to quantify the continuum of each spectrum. These laboratory analyses will contribute to the understanding of how factors including regolith porosity and albedo change the resulting spectra. Such insights will be instrumental in interpreting telescopic observations of asteroid bodies, akin to those obtained from the James Webb Space Telescope (JWST).

## Acknowledgments & Citations

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Martin, A. C., Emery, J. P., and Loeffler, M. J., "Spectral effects of regolith porosity in the mid-IR - Forsteritic olivine", *Icarus*, vol. 378, 2022. doi:10.1016/j.icarus.2022.114921.